

相对速度 $U = v_0 + v$ 随颗粒减速而改变

$$v\,dv/dz = -\eta\rho_0(v_0+v)^2/(\rho r)$$

$$s = \rho r(\ln 2 - 1/2)/(\eta\rho_0)$$

恒定 $U = 2v_0$ 另列为初值估算